

# **PSYCHOLOGY AND NEUROSCIENCE 49<sup>TH</sup> ANNUAL GRAHAM GODDARD IN-HOUSE CONFERENCE**



**June 20, 2025  
Life Sciences Centre  
Rooms 4260 and 5260**

## IN-HOUSE SCHEDULE – FRIDAY, JUNE 20<sup>TH</sup>

### TALKS

(All talks will be held in LSC Psychology, rooms 4260 & 5260 on the 4<sup>th</sup>-5<sup>th</sup> floor)

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#### 9:00-10:15 Session 1 (Chair: Ray Klein) LSC Psychology, Room 4260 on the 4<sup>th</sup> floor – *The 50 Years Symposium: To Celebrate 50 years Since the First In-House Conference, by Some Who Have Been Here for 50 Years*

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- 9:00 Aaron Newman *Opening Remarks*  
T1) 9:15 Ray Klein *50 Years in Psychology (and Neuroscience) at Dalhousie*  
T2) 9:35 Donald Mitchell *The Kitten Breeding Colony in the Psychology Department: The First 50 Years*  
T3) 9:55 Richard Brown *Chemical Signals, Olfactory Learning and Individual Differences: 50 Years of Research*

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#### 10:15-10:45 Coffee Break - 4<sup>th</sup> Floor Lilyan E. White Study Centre

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#### 10:45-12:30 Session 2 (Moderator: Sherry Stewart) LSC Psychology, Room 4260 on the 4<sup>th</sup> floor – *Studies of Substance Use Patterns, Cue-Induced Craving, Cross-Priming, and Expectancy Effects: The Wide-ranging Behavioral Psychopharmacology Legacy of Dr. Sean Barrett*

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- T4) 10:45 Jessica Peterson *Examining Patterns of Substance Use Among Simultaneous Alcohol and Cannabis Co-Users*  
T5) 11:00 Robin Perry *Menthol Enhances Sensory Reinforcement and Craving Reduction from Nicotine Inhalers*  
T6) 11:15 Kim Good *Smoking Cue Reactivity and Smoking Motives in Patients with Early Phase Psychotic Disorder: A Behavioural Study*  
T7) 11:30 Sherry Stewart *Cross-Priming of Stimulant Cravings by Gambling in Regular Stimulant Users*  
T8) 11:45 Amelie Tanner *It's About Time: Distinguishing the Influence of Reflexive and Volitional Mechanisms of Temporal Attention*  
T9) 12:00 Mikaela Ethier-Gagnon *The Effects of Cannabidiol (CBD) Expectancy on Subjective Stress, Anxiety, and Intrinsic Connectivity Network Functional Connectivity in Healthy Adults*  
T10) 12:15 Igor Yakovenko *Discussant*

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#### 12:30-14:00 Lunch & Poster Session - 4<sup>th</sup> Floor Lilyan E. White Study Centre

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#### 14:00-15:00 Session 3A (Chair: Sean Mackinnon) LSC Psychology, Room 4260 on the 4<sup>th</sup> floor

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- T11) 14:00 Holly Carbyn *Evaluating the Impact of an Online Coaching Intervention for Families Affected by 22q11.2 Deletion Syndrome*  
T12) 14:15 Nick Murray *Objective Behavioural Comparison of Youth and Adult Anxiety: A Mobile Sensing Approach*  
T13) 14:30 Alexander Farquhar *The Relationship of Perfectionism, Culture, Acculturative Stress and Depression in University Students*  
T14) 14:45 Sean Mackinnon *Comparing Structural Equation Models to Network Analysis Models of Depression, Anxiety, and Perfectionism*

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**14:00-15:00 Session 3B (Chair: Brodie Badcock-Parks) LSC Psychology, Room 5260 on the 5<sup>th</sup> floor**

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- T15) 14:00 **Mackenzie Hartley** *The Impacts of Chronic Late Adolescent Stress on Adult Sociability in Mice*  
T16) 14:15 **K. C. Collings** *Brains in Tune with Season: Seasonal Regulation of Neurogenesis Within Vocal Control and Auditory Perceptual Regions of Wild Black-capped Chickadees (*Parus atricapillus*)*  
T17) 14:30 **Hannah Miller** *Volumetric Plasticity of the Septum Across Age and Feeding Conditions in Domesticated Pigeons (*Columba Livia Domestica*)*  
T18) 14:45 **Brodie Badcock-Parks** *Ageing, Metabolic Challenge, and Neural Plasticity: Transcriptomic Insights from Domesticated Pigeons (*Columba Livia Domestica*)*

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**15:00-15:15 Coffee Break - 4<sup>th</sup> Floor Lilyan E. White Study Centre**

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**15:15-16:15 Session 4A (Chair: Gail Eskes) LSC Psychology, Room 4260 on the 4<sup>th</sup> floor**

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- T19) 15:15 **Ralph Redden** *Standing up in the Virtual World: Observing Social-Cognitive Factors Influencing Peer Defending Behaviours Using Virtual Reality*  
T20) 15:30 **Bei Ni Tan** *Improving Functional Connectivity Estimation in rs-fMRI for Biomarker Discovery Across Neurological Disorders*  
T21) 15:45 **Lucas Tennen** *Developing an Online Treatment Approach for Spatial Neglect Using a Single Case Design*  
T22) 16:00 **Richard Drake** *Content Variability as Desirable Difficulty in N-Back Training*

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**15:15-16:30 Session 4B (Chair: Aaron Newman) LSC Psychology, Room 5260 on the 5<sup>th</sup> floor**

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- T23) 15:15 **Wateen Sadek** *Brains in Conversation: Science Communication and Public Impact*  
T24) 15:30 **Can Sozuer** *Using AI to Detect AI Use in Online Exams*  
T25) 15:45 **Yao Li** *Cerebellar Isolation Combining Multi-Modality MRI Using Deep Learning*  
T26) 16:00 **Madison MacLachlan** *The Impact of Language Background and Prosocial Behaviours on Preschoolers' Affiliation Preference Towards Unfamiliar Accented Speakers*  
T27) 16:15 **Hope Lewis** *Changes in the N400 Event-Related Potential Found Over a Two Day Spanish Language Learning Study*

## POSTERS

(All posters will be presented from 12:30 pm to 2 pm in the Lilyan E. White Study Centre on the 4<sup>th</sup> floor)

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|----|---------------------|----|-----------------|
| 1  | Mohammed Ali Ahmend | 12 | Salma Ismail    |
| 2  | Isabel Atwell       | 13 | Wes Jones       |
| 3  | Varun Cairae        | 14 | Kirsty Longino  |
| 4  | Mya Dockrill        | 15 | Erin MacDougall |
| 5  | Ben Drew            | 16 | Samantha Savoie |
| 6  | Ana Faustova        | 17 | Can Sozuer      |
| 7  | Oswin Fernandes     | 18 | Mark ten Haaf   |
| 8  | Brett Feltmate      | 19 | Caden Whitfield |
| 9  | Sarah Haggett       | 20 | Maura Whitman   |
| 10 | Aurthy Imam         | 21 | Deniz Yildirim  |
| 11 | Bronwyn Inness      |    |                 |

## TALK ABSTRACTS

### Badcock-Parks, Brodie

**TALK TITLE:** Ageing, metabolic challenge, and neural plasticity: transcriptomic insights from domesticated pigeons (*Columba livia domestica*)

**AUTHORS:** Broderick M. B. Parks<sup>1\*</sup>, T. Adam Liddle<sup>2</sup>, Tyler J. Stevenson<sup>2</sup>, Jason R. Treberg<sup>3</sup>, Debbie M. Kelly<sup>4</sup>, Leslie S. Phillimore<sup>1</sup>

<sup>1</sup>Department of Psychology and Neuroscience, Dalhousie University, Halifax NS, Canada

<sup>2</sup>School of Biodiversity, One Health, and Veterinary Medicine, University of Glasgow, Glasgow, Scotland, United Kingdom

<sup>3</sup>Department of Biological Sciences, University of Manitoba, Winnipeg MB, Canada

<sup>4</sup>Department of Psychology, University of Manitoba, Winnipeg MB, Canada

\*Presenting author

**ABSTRACT:** Domesticated pigeons (*Columba livia domestica*) are valuable models for studying spatial cognition and neural plasticity. Unlike mammals, birds exhibit extensive plasticity throughout adulthood, but it remains unclear whether the avian brain shows any evidence of (mammal-like) decline with age. While food deprivation is often used to experimentally motivate spatial behaviour, its effects across different life stages are also unclear. Physiological differences between adult and aged birds suggests a role for the mediobasal hypothalamus, a key regulator of energy balance, but whether age-related physiological differences (and/or responses to food deprivation) reflect molecular changes in this structure remains unexplored. Here, we examined how a standard 90% food deprivation affected neural plasticity in adult (1-4 years) and aged (8-18 years) pigeons. Birds ( $N=24$ ) were assigned to one of three food conditions over ~16 weeks: (i) recent food deprivation; (ii) past food deprivation; or (iii) *ad libitum* feed. Using Nanopore RNA sequencing, we profiled all RNA transcripts in the mediobasal hypothalamus and hippocampus, identifying hundreds of differentially expressed genes based on age, food condition, and their interaction. These findings reveal previously unknown molecular signatures of aging and metabolic challenge in the avian brain, providing new insights into neural plasticity and energy regulation across the lifespan.

### Brown, Richard

**TALK TITLE:** Chemical signals, olfactory learning and individual differences: 50 years of research. [Chemical Signals in Vertebrates 16. Edited by B. Schaal et al. In Press, May 2025.]

**AUTHORS:** Richard E. Brown

**ABSTRACT:** This talk describes 50 years of research in my lab on chemical signals in rodents. These began with experiments on the role of odour preferences and urine-marking in the socio-sexual behaviour of rats. Olfactory communication involves both endocrine and experiential factors as shown by our studies on gonadectomy and isolation rearing. The importance of individual recognition by odours was studied through the rat pups' response to "maternal pheromones" and by examining the role of the major histocompatibility complex in odours of individuality. The initial studies done using habituation-dishabituation tests were continued using an operant olfactometer to study odour discrimination learning using the urine odours of rats and mice. We also investigated the importance of individual recognition in puberty acceleration in juvenile female mice. We then began to examine the use of the olfactometer to study working memory in rats and learning set formation in mice. We are now examining olfactory reversal learning in mice using signal detection methods to identify sex, age and genotype differences in learning strategies. We also use the Pavlovian conditioned odour preference task developed by Heather Schellinck to study olfactory learning and memory. Our studies have attempted to integrate mechanisms of olfactory communication (releaser effects), neuroendocrine responses (primer effects) and learning and memory (cognitive processes) in olfactory signalling in rodents.

## Carbyn, Holly

**TALK TITLE:** Evaluating the Impact of an Online Coaching Intervention for Families Affected by 22q11.2 Deletion Syndrome

**AUTHORS:** Holly Carbyn Abiaz Hossain, Patricia Lingley-Pottie, Lisa D. Palmer, Sophie Ayoub, Andrea Shugar, Donna M. McDonald-McGinn, Patrick J. McGrath, Anne S. Bassett, Cheryl Cytrynbaum, Ann Swillen & Sandra Meier.

**ABSTRACT:** 22q11.2 Deletion Syndrome (22q11DS) is the most common microdeletion syndrome, with broad phenotypic variability often involving high levels of internalizing/externalizing symptoms in children. Caregivers of children with 22q11DS are at increased risk of poor mental health, yet caregiver-focused supports remain scarce. This pilot study evaluated the effectiveness of an online coaching intervention for families affected by 22q11DS using a repeated-measures design, examining its impact on caregiver mental health, parenting practices, and child internalizing/externalizing symptoms. Eighteen caregivers completed the intervention, which included 12 weekly modules with coached group phone calls. Outcomes were assessed at baseline, 3-, and 6-month follow-up using the DASS-21 (parent distress), PAFAS (parenting practices), and SDQ (child symptoms). Linear mixed models revealed significant reductions in caregiver stress at both follow-ups and a marginal decrease in anxiety at 6 months. Children's externalizing behaviours declined significantly over time. Findings suggest the online intervention may alleviate caregiver distress and improve child behaviour in families affected by 22q11DS. Larger studies (RCTs) are warranted to confirm results.

## Collings, K. C.

**TALK TITLE:** Brains in Tune with Season: Seasonal Regulation of Neurogenesis Within Vocal Control and Auditory Perceptual Regions of Wild Black-capped Chickadees (*Poecile atricapillus*)

**AUTHORS:** KC Collings<sup>1,2,a</sup>, Broderick M. B. Parks<sup>2</sup>, Leslie S. Phillmore<sup>2</sup>

<sup>1</sup>Department of Biology, Dalhousie University, Halifax, NS, Canada

<sup>2</sup>Department of Psychology and Neuroscience, Dalhousie University, Halifax, NS, Canada

<sup>a</sup>Current affiliation: Department of Biology, Saint Mary's University, Halifax, NS, Canada

**ABSTRACT:** Songbird vocal behaviour often changes seasonally, with birds singing more in the spring to attract mates and defend territories. This behavioural plasticity is supported by neural plasticity specifically within the vocal control system, the interconnected regions and circuits supporting songbird vocal behaviour. One measure of neural plasticity is neurogenesis: the proliferation, differentiation, migration, and incorporation of new neurons. Here, we evaluated seasonal differences in neurogenesis within four brain regions: two vocal control nuclei (HVC [proper name] and Area X) and two auditory perceptual regions (caudomedial nidopallium and caudomedial mesopallium) in adult male and female black-capped chickadees (*Poecile atricapillus*). We captured birds across three seasons (spring, summer, and winter) and quantified neurogenesis by labelling brain tissue for doublecortin (DCX), a marker of newborn neurons. We counted three morphologies of DCX+ cells (multipolar, fusiform, and round) and found the distribution of these morphologies differed in each brain region. We also found seasonal changes in HVC multipolar cells, and sex-specific seasonal differences in Area X round cells. These results suggest that seasonal patterns of neurogenesis relate to the functional roles of the different brain regions and also vary between sexes, broadening our overall understanding of avian physiology, behaviour, and neural plasticity.

## Drake, Richard

**TALK TITLE:** Content Variability as Desirable Difficulty in N-Back Training

**AUTHORS:** Richard S. Drake, Sophie Bhaskara, Raymond M. Klein, & Gail A. Eskes

**ABSTRACT:** Working memory training research has been a grand quest for transfer (improvements on related but untrained tasks). Recent culture shifts in the field have reprioritized mechanistic studies probing transfer theories, and one underexplored area involves manipulations of training variability. Our VARIA project is a 5-day training study that draws together multiple learning theories, from Schmidt and Bjork's work on desirable difficulties in motor and

vocabulary learning to Gathercole's theory of novel subroutines as transferable units of cognitive skill learning. We embedded a categorization task within a traditional n-back paradigm to create a novel "varback" training program that we used to test the hypothesis that training with high variability (3x as much as a lower variability group) will boost retention and near transfer measures of working memory. We will present and contextualize preliminary findings with respect to contemporary debates in working memory training.

### Ethier-Gagnon, Mikaela

**TALK TITLE:** The Effects of Cannabidiol (CBD) Expectancy on Subjective Stress, Anxiety, and Intrinsic Connectivity Network Functional Connectivity in Healthy Adults

**AUTHORS:** Ethier-Gagnon, M. A., Perry, R.N., Helmick, C., Crocker, C., Good, K., Stewart, S. H., & Barrett, S. P.

**ABSTRACT:** Cannabidiol (CBD) may be a promising candidate for stress/anxiety disorder treatment. CBD expectancy alone impacts subjective, physiological, and endocrine markers of stress/anxiety. We aimed to delineate CBD expectancy effects on subjective state and resting state functional connectivity (rsFC) within and between the default mode network (DMN), salience network (SN), and central executive network (CEN). Healthy adults (N=32) were randomized to receive accurate or misleading instructions regarding the CBD content of a CBD-free oil. rsFC was assessed at baseline and following a stress task. Subjective state was measured at multiple timepoints. In the Told CBD-free (control) condition, the SN showed increased rsFC with both the CEN and DMN. In the Told CBD (expectancy) condition, decreases in stress and anxiety were reported following oil administration and during stressor recovery, and increased rsFC was observed within and between the DMN and CEN. Results suggest CBD expectancy may engage neural pathways involved in stress-regulation.

### Farquhar, Alexander

**TALK TITLE:** The Relationship of Perfectionism, Culture, Acculturative Stress and Depression in University Students

**AUTHORS:** Alexander Farquhar

**ABSTRACT:** The stress generation hypothesis suggests that perfectionistic tendencies can create stress, increasing the risk of depression. This may apply to international students who experience acculturative stress, defined as a stress reaction towards struggles associated with adapting to a different culture. This study investigated whether acculturative stress mediates the relationship between perfectionism and depression. Participants were 119 Dalhousie University students who identified as international students or first-generation immigrants. They completed an online survey including measures of perfectionism (Big Three Perfectionism Scale), acculturative stress (Riverside Acculturative Stress Inventory), and depression (Center for Epidemiologic Studies Depression Scale). Mediation analysis revealed no indirect effect for rigid or self-critical perfectionism. However, narcissistic perfectionism was found to indirectly predict depression through acculturative stress. When controlling for time in Canada, this indirect effect was no longer significant. These findings highlight a potential link between narcissistic perfectionism and stress generation in the context of cultural adjustment and suggest that acculturative stress may be a pathway to depression for this perfectionism subtype.

### Good, Kim

**TALK TITLE:** Smoking Cue Reactivity and Smoking Motives in Patients with Early Phase Psychotic Disorder: A Behavioural Study

**AUTHORS:** Good, K., Couture, M.-E., Stewart, S. H., Barrett, S. P., Tibbo, P., Milliken, H., Alexiadis, M., & Robertson, N.

**ABSTRACT:** Rates of tobacco smoking and unsuccessful smoking cessation attempts are high in individuals with psychosis. Exposure to smoking-related cues reliably enhances cigarette cravings in dependent smokers. Smoking-related cues do not affect smokers with chronic schizophrenia to the same degree as for non-psychotic smokers. However, less is known about individuals in the early phase of psychotic (EPP) illness. Additionally, it is unknown whether the smoking motives of EPP differ from those of non-psychiatric smokers. Cue-elicited cigarette cravings, and self-reported motives for smoking,



were compared between EPP individuals (n=21) and non-psychiatric participants (NC; n=40). EPP individuals were less sensitive than NC to the cue-reactivity paradigm, suggesting individuals in the early phase of illness may not be affected by tobacco cues to the same degree as non-psychiatric smokers. Further, smoking motives did not differ between EPP and non-psychiatric smoker controls. Findings have implications for differences in smoking cessation interventions for these two populations.

### Hartley, Mackenzie

**TALK TITLE:** The Impacts of Chronic Late Adolescent Stress on Adult Sociability in Mice

**AUTHORS:** Mackenzie R. Hartley, Caden Whitfield, Pouria Vallapour Karimi, Tamara B. Franklin

**ABSTRACT:** Adolescents are more sensitive to stress than adults and experiencing stress during this developmental period can have lasting effects on behaviour, including increased depression-like and anxiety-like behaviours. However, its long-term impact on sociability has not been well studied. The present research aims to determine the effects of chronic late adolescent stress on adult peer-to-peer sociability using male and female, stressed and control mice. Male and female mice (n=48) were evenly split into stress and control groups. From post-natal day (PND) 48-60, we exposed stressed mice to a chronic predator stress paradigm while controls were left undisturbed in their home cages. Once mice reached adulthood (PND 96), we performed behavioural testing aimed at quantifying social dominance and social interest in same-sex and opposite-sex peers. Results suggest that chronic adolescent stress may lead to increased interest in same-sex peers and decreased social dominance. These findings could inform research on the lasting effects of human childhood and adolescent chronic stressors such as unsafe home environments and mental health struggles.

### Klein, Ray

**TALK TITLE:** 50 years in Psychology (and Neuroscience) at Dalhousie

**AUTHORS:** Raymond Klein

**ABSTRACT:** My time at Dalhousie coincides with the history of the Department's annual, and widely copied, In-House Conference. I plan to entertain you with facts, figures and stories that cover three histories: the Department's, the In-House's and mine.

### Lewis, Hope

**TALK TITLE:** Changes in the N400 Event-Related Potential Found Over a Two Day Spanish Language Learning Study

**AUTHORS:** Hope Lewis, Emily Wei, Mobina Ali, Aaron Newman

**ABSTRACT:** The N400 is an event-related potential (ERP) associated with semantic processing. Before training in an unfamiliar language (UL), learners do not show this N400 effect when assessed with UL words, but such effects have been shown to emerge at least after 10 days of training. To determine how much exposure to UL words are required for new learners to show an N400 effect, the current study recorded EEG throughout a two-day UL training period. Spanish words were taught by being paired with images associated with their meaning. Semantic mismatch tasks were used to elicit ERPs in Spanish and in English. A significant N400 effect for trained Spanish words emerged at the end of day 2 of training. This N400 effect was delayed and smaller in amplitude compared to English, perhaps reflecting a difference in proficiency. Follow-up study will assess participants over a semester of language learning to determine longer term development.



## Li, Yao

**TALK TITLE:** Cerebellar Isolation Combining Multi-Modality MRI Using Deep Learning

**AUTHORS:** Yao Li<sup>1</sup>, Jorn Diedrichsen<sup>2</sup>, Carlos Hernandez-Castillo<sup>1</sup>

1. Faculty of Computer Science, Dalhousie University
2. Brain and Mind Institute, Western University

**ABSTRACT:** The cerebellum is involved in motor, cognitive and social behavior, and understanding its development across the lifespan aids in diagnosing neurological disorders. Accurate cerebellum isolation from whole-brain MRI scans is a crucial first step for accurate cerebellar analysis. The current isolation tools have two limitations. First, they were developed on healthy adults' brains. Second, most of them do not utilize multi-modality image inputs. This project introduces a robust deep learning approach using a 3D U-Net architecture to isolate the cerebellum from multi-modal MRI data (T1- and T2-weighted images). The dual-channel input enhances tissue differentiation by capturing spatial features from both modalities. Evaluated using Dice Score and Hausdorff Distance, our model outperforms existing tools in accuracy, particularly by reducing mislabeling of nearby structures. It also generalizes well across different age groups, including neonates, addressing key limitations of previous methods and offering a more reliable solution for cerebellar segmentation.

## Mackinnon, Sean

**TALK TITLE:** Comparing Structural Equation Models to Network Analysis Models of Depression, Anxiety, and Perfectionism

**AUTHORS:** Sean P. Mackinnon & Andy J. Kim

**ABSTRACT: Background:** We use network analysis to identify bridge symptoms connecting anxiety, depression and perfectionism; that is, are there specific symptoms connecting these constructs? Moreover, we compare network models to traditional structural equation models using latent variables to see how findings differ.

**Methods:** Two undergrad samples (N=774; N=759) completed online measures of anxiety, depression, and perfectionism (concerns over mistakes, doubts about actions, & personal standards).

**SEM Results:** After controlling for other perfectionism variables, concern over mistakes and doubts about actions were positively associated with depressive and anxiety symptoms. Personal standards were negatively associated with depression and non-significantly related to anxiety.

**Network Results:** Four items emerged as bridge symptoms: difficulty taking initiative to do activities, feeling worthless, feeling close to panic, and doubts about simple everyday activities.

**Conclusions/Impact:** Traditional multiple regression-style models are replicable but obscure item-specific relationships. Using network models in place of reflective latent variable models may produce novel insights.

## MacLachlan, Madison

**TALK TITLE:** The Impact of Language Background and Prosocial Behaviours on Preschoolers' Affiliation Preference Towards Unfamiliar Accented Speakers

**AUTHORS:** Madison B. MacLachlan, Dr. Drew Weatherhead

**ABSTRACT:** Preschoolers prefer people who speak with the same accent as them (e.g., Kinzler, et. al., 2019) and who behave kindly towards others (e.g., Van de Vondervoort & Hamlin, 2017). Recently, it has been found that a preschoolers' language background (i.e., whether they are monolingual or multilingual) may impact these preferences. Bilinguals are more flexible in their preferences based on language (e.g., Byers et al., 2017) and show more sensitivity to kindness cues (Singh, 2020). The present study examined the impact of language background and social behaviours on preschoolers' preferences towards animated characters with an unfamiliar accent. N= 158 preschoolers answered

questions regarding their preferences toward the group. Participants significantly preferred prosocial and neutral characters over antisocial characters and bilinguals showed significantly greater affiliation preference towards the accented characters compared to the monolinguals. This suggests that bilinguals may be more accepting socially of speakers with accents different than their own.

### Millar, Hannah

**TALK TITLE:** Volumetric Plasticity of the Septum Across Age and Feeding Conditions in Domesticated Pigeons (*Columba livia domestica*)

**AUTHORS:** Hannah Millar<sup>1,2</sup>, Broderick M. B. Parks<sup>2</sup>, Casandra Hayar<sup>2</sup>, Ainsley N. Monchesky<sup>2</sup>, Jason R. Treberg<sup>3</sup>, Debbie M. Kelly<sup>4</sup>, Leslie S. Phillmore<sup>2</sup>

<sup>1</sup>Department of Biology, Dalhousie University, Halifax, NS, Canada

<sup>2</sup>Department of Psychology and Neuroscience, Dalhousie University, Halifax, NS, Canada

<sup>3</sup>Department of Biological Sciences, University of Manitoba, Winnipeg, MB, Canada

<sup>4</sup>Department of Psychology, University of Manitoba, Winnipeg, MB, Canada

**ABSTRACT:** Domesticated pigeons (*Columba livia domestica*) are common models to investigate the interaction between spatial memory and its underlying neural substrates, particularly the hippocampus. Relative to the hippocampus, much less is known about the avian septum, another structure that has been linked to general spatial specialization in birds, is highly interconnected with the hippocampus, and shows some evidence of neural plasticity. Here, I will present data from two ongoing studies examining the effects of age, sex, and food deprivation history on neural plasticity in the pigeon septum. The first study examined age-related changes in overall septum volume (as measured by Nissl staining) in pigeons ( $N=30$ ) ranging in age from 2-22 years old. The second study examined how food deprivation affected septum volume in a separate sample ( $N=22$ ) of adult (1-4 years) and aged (8-18 years) pigeons. While overall telencephalon volume varied with both age and food condition, we found no effects on septum volume. We did find some evidence of septum plasticity in response to food deprivation in adult, but not aged birds, however further methodological refinements may be necessary to confirm this effect. Overall these findings suggest the septum may be resilient to potentially detrimental effects of age and metabolic challenge.

### Mitchell, Donald

**TALK TITLE:** The kitten breeding colony in the Psychology Department: the first fifty years.

**AUTHORS:** Donald Mitchell

**ABSTRACT:** The establishment of a kitten breeding colony at Dal was a crucial initial step in order to study of the role that early visual experience plays in the development of the mammalian central visual pathways of binocular species such as humans. As described in a recent paper (<https://doi.org/10.1523/ENEURO.0576-24.2025>) it was also necessary to develop a method to allow measurement of the changes in the perceptual abilities of kittens following different forms of early visual exposure. In 1970 there were no commercial cat breeding colonies in Canada and we had not yet moved into the LSB. I describe how and from where we acquired intact cats for the initiation of a breeding colony and the progressive changes made in the ensuing years.

### Murray, Nick

**TALK TITLE:** Objective Behavioural Comparison of Youth and Adult Anxiety: A Mobile Sensing Approach

**AUTHORS:** Nicholas E. Murray, Mah Parsa, Mohamed Muzamil, JianLi Wang, Sandra M. Meier

**ABSTRACT:** Widely used subjective measures of anxiety are limited by recall biases and an inability to capture real-time fluctuations in anxiety-related behaviours. This study used the PROSIT app to examine behavioural indicators of anxiety

in 65 youth (15-21 years) and 96 adults (26-40 years) with clinically significant anxiety. Participants completed baseline measures of mental health and used the app for 14 days, enabling passive sensing of mobility, sociability, physical activity, screen use, and sleep, alongside daily self-reports. Features were extracted to predict outcomes related to baseline and momentary anxiety. Notably, screen use predicted anxiety in youth only; greater mobility predicted greater momentary anxiety variance in youth, and lesser variance in adults. These results— along with other findings to be discussed – indicate that mobile sensing features can predict anxiety in youth and adults, and that there are measurable differences in the expression of anxiety between youth and adults consistent with prior literature.

### Perry, Robin

**TALK TITLE:** Menthol Enhances Sensory Reinforcement and Craving Reduction from Nicotine Inhalers

**AUTHORS:** Perry, R.N., Tanner A., van Staden, F., O'Connor, G., & Barrett, S.P.

**ABSTRACT:** Flavours, particularly menthol, may influence the reinforcing properties of electronic nicotine delivery systems (ENDS), though findings in human users are mixed. We examined whether menthol alters reinforcement from nicotine inhalers in adult ENDS users. Fifty-three ENDS users self-administered a 4mg nicotine inhaler paired with a menthol or unflavoured inhaler across two within-subject sessions. Craving and withdrawal were assessed pre- and post-inhaler; subjective effects (liking, wanting, stimulation, sensory effects) were rated post-inhaler. Craving and withdrawal decreased pre- to post-inhaler in both flavour conditions ( $p < 0.001$ ). Planned comparisons between flavours showed lower post-inhaler craving in the menthol condition ( $p = 0.04$ ). Menthol also elicited cooling sensory effects, improved the taste of nicotine inhalers, and independently increased ratings of stimulation, liking, and wanting ( $p < .001$ ). Thus, menthol may enhance the sensory appeal and reinforcing effects of inhaled nicotine, including craving reduction. Findings support regulatory concerns about flavoured ENDS while highlighting nuanced implications for cessation and harm reduction.

### Peterson, Jessica

**TALK TITLE:** Examining Patterns of Substance Use Among Simultaneous Alcohol and Cannabis Co-Users

**AUTHORS:** Peterson, J., Shephard A., Ethier-Gagnon M. A., Zhevoka, R., & Barrett S. P.

**ABSTRACT:** Simultaneous co-use of alcohol and cannabis has been associated with greater substance use and dependence relative to concurrent or single substance use. We examined use patterns among simultaneous co-users to identify factors that may contribute to this elevated risk. Structured interviews ( $n = 200$ ) were conducted in monthly alcohol-cannabis co-users. General linear models were used to analyze differences in alcohol and cannabis dependence between concurrent and simultaneous users, and to examine how specific use patterns influenced substance quantity among simultaneous users. Relative to concurrent users, simultaneous users demonstrated more problematic alcohol and cannabis use. Among simultaneous users, order in which alcohol and cannabis were administered impacted quantity used, such that the substance administered first was used more in the session. Finally, we found that more alcohol was consumed in group vs. solitary settings. Findings highlight key risk factors associated with simultaneous co-use and may inform targeted psychoeducation and harm-reduction strategies.

### Redden, Ralph

**TALK TITLE:** Standing up in the Virtual World: Observing Social-Cognitive Factors Influencing Peer Defending Behaviours Using Virtual Reality

**AUTHORS:** Julia Byron, Anna MacGillivray, Laura Lambe and Ralph Redden

**ABSTRACT:** Bullying victimization is a serious threat to Canadian youth. Bullying most often occurs in the presence of peer bystanders, who can intervene to defend their victimized peers. Although rare, peer defending is an effective way to combat bullying, ending the bullying episode rapidly when a bystander intervenes. Past research has used passive viewing methods (e.g. vignettes) to study factors associated with peer defending, limiting our understanding of how social attention contributes to peer defending in real-time. The present study used a virtual reality (VR) paradigm to

examine how social attention is related to peer defending in a bullying scenario. Multi-method data were collected during the VR task, including eye tracking data, state affect, and various defending behaviours in response to a social exclusion scenario. Results indicate that there is a correlation between gaze allocation and some types of peer defending, as well as a correlation between specific peer defending behaviours and negative affect during social exclusion. Results from this study are among the first to demonstrate the social cognitive processes underlying peer defending behaviour as it unfolds in real-time, showing that gaze allocation in dynamic social situations can be sensitive to witnessing peer victimization.

### Sadek, Wateen

**TALK TITLE:** Brains in Conversation: Science Communication and Public Impact

**AUTHORS:** Wateen Sadek

**ABSTRACT:** This presentation explores the role of science communication in making neuroscience more accessible and engaging to the public. Drawing on my involvement with outreach initiatives such as SURGE, Dalhousie Science Scholars and Science Leaders Program (DSSLP), the Brain Repair Centre, and high school workshops, I highlight how interactive demonstrations using EEG and brain-computer interface (BCI) technology can spark curiosity and enhance science literacy among young people. As a neuroscience major with minors in entrepreneurship and education, I aim to bridge the gap between complex scientific research and community understanding. The presentation also reflects on the impact of hands-on activities and youth-centred programming in fostering future interest in neuroscience. These experiences demonstrate how science communication can expand the reach of research, inspire young minds, and foster stronger public engagement with neuroscience.

### Sozuer, Can

**TALK TITLE:** Using AI to Detect AI Use in Online Exams

**AUTHORS:** Can Sozuer

**ABSTRACT:** Unproctored online examinations are highly susceptible to covert artificial-intelligence (AI) use and peer collaboration, undermining assessment integrity. Although manual review can uncover isolated incidents, its scalability is prohibitive. To bridge this gap, I developed Zapper, a custom AI-based advanced data analysis tool that processes textual student responses to flag question- and student-level signatures of AI usage and collaboration. For a class of ~300 students, Zapper reduces ~8,000 person-hours of manual inspection into a review of its outputs and student work lasting only a few hours. Applied over three semesters in two distinct courses with close to 700 students in total, Zapper consistently achieved high accuracy. By integrating advanced AI-detection capabilities into a scalable framework, Zapper advances the integrity of unproctored online assessments.

### Stewart, Sherry

**TALK TITLE:** Cross-Priming of Stimulant Cravings by Gambling in Regular Stimulant Users

**AUTHORS:** Stewart, S. H., Zack, M., McGrath, D., Seale, I., Collins, P., Barrett, S. P., & Al-hamdani, M.

**ABSTRACT:** Administration of addictive drugs can prime motivation to use other drugs with similar properties (i.e., “cross-priming”; Barrett et al., 2006). Amphetamine administration increases urge to gamble in problem gamblers (Zack & Poulos, 2004). We tested whether motivation for stimulant use would be similarly cross primed by gambling in regular stimulant users. Stimulant users (n=32) and n=31 controls (n=15-16 of each sex/group): engaged in lab-based video lottery terminal gambling; reported gambling, stimulant, and alcohol craving pre- and post-gambling; and provided heart rate and blood pressure samples at both timepoints. Gambling increased desire to gamble. While heart rate and systolic blood pressure declined pre- to post-gambling in controls, they remained elevated post-gambling in stimulant users. Gambling also cross-

primed increased craving for stimulants and alcohol but only in the female stimulant users. Results bolster arguments that problem gambling shares common mechanisms with psychostimulant addiction and suggest cross-priming may be particularly important in females.

### Tan, Bei Ni

**TALK TITLE:** Improving Functional Connectivity Estimation in rs-fMRI for Biomarker Discovery Across Neurological Disorders

**AUTHORS:** Bei Ni Tan<sup>1</sup>, Carlos Hernandez-Castillo<sup>1</sup>

1. Faculty of Computer Science, Dalhousie University

**ABSTRACT:** Neurological disorders, caused by central and peripheral nervous system abnormalities, often impair brain function. Functional connectivity (FC) from Resting-state functional MRI (rs-fMRI) is a promising tool for biomarker discovery in neurological disorders. However, current methods to calculate FC may overlook subtle connectivity patterns leading to weak predictive power. Using three different rs-fMRI datasets, we evaluate different FC estimation methods: Pearson, Spearman, empirical covariance, Ledoit-Wolf shrinkage, and oracle approximation shrinkage. Then we train six machine learning classifiers to evaluate their performance on identifying neurological disorders. Feature importance scores from the best classifiers were used to identify disorder-relevant FC patterns. Results show that shrinkage-based methods outperform traditional correlation approaches, and our proposed weighted connectivity matrix combined with feature importance analysis enhances classification. Linear SVM and logistic regression achieved the highest accuracy. These findings demonstrate the potential of machine learning to improve FC estimation and support biomarker discovery across neurological disorders.

### Tanner, Amelie

**TALK TITLE:** Cannabidiol Expectancy Attenuates Users' Anxiety Response to Cannabis Cues

**AUTHORS:** Tanner, A. B., Shephard, A. S., Barrett, S. P., Sherry, S., & Stewart, S. H.

**ABSTRACT:** Cannabidiol (CBD) dampens psychophysiological arousal and substance craving; however, the extent to which these effects are attributable to placebo effects remains unclear. CBD expectancy alone has been shown to elicit anxiolysis and attenuate cue-induced cannabis craving. We examined whether CBD expectancy influences stress or anxiety responses to cannabis-related cues. Forty-six cannabis users self-administered a vaporizer containing propylene glycol; half were falsely informed that the vaporizer contained CBD. Participants then viewed neutral and cannabis-related cues. Stress and anxiety were assessed at baseline, following vaporizer administration, and after each cue. Participants who believed they received CBD showed significant reductions in anxiety following drug (but not neutral) cue exposure compared to baseline, while those who believed they received placebo maintained higher anxiety ratings post-drug cue. Findings suggest CBD expectancy alone is sufficient to dampen anxiety to cannabis-related cues, underscoring the need to control for expectancy effects in clinical research investigating CBD's therapeutic efficacy.

### Tennen, Lucas

**TALK TITLE:** Developing an Online Treatment Approach for Spatial Neglect Using a Single Case Design

**AUTHORS:** Lucas Tennen, BA1, Nick Murphy, BSc2, Samantha Horne, BSc1, Julia DuJohn1, Joy Boyce, BSc (OT)3, Sherri Carter, PhD3, Anita Mountain, MD3, Raymond M. Klein, PhD1, Gail Eskes, PhD1.

1 Dalhousie University, Halifax, NS; 2 McGill University, Montreal, QC; 3 Nova Scotia Health, Halifax, NS.

**ABSTRACT:** Spatial neglect, a common consequence of right hemisphere stroke, leads to attentional deficits and poor rehabilitation outcomes. While visual scanning treatment (VST) is a potential intervention, its effectiveness typically depends on therapist-guided sessions exceeding 15 hours, posing challenges in resource-limited settings. SpyTrain, an online VST program, offers intensive, adaptive, game-like scanning practice and has shown validity in healthy controls (Aziz et al., 2021), and early efficacy in stroke patients. This study is assessing the feasibility and effectiveness of SpyTrain

for treating spatial neglect, focusing on the application of strategies. Patients with right hemisphere stroke and neglect are randomized to three different baseline lengths using a non-concurrent multiple baseline single-case design, followed by 5 weeks of online VST (5 sessions/week). Outcomes will be measured across phases and analyzed using Tau-U, a nonparametric test assessing improvement trends. If effective, SpyTrain could offer an accessible, home-based treatment to support recovery and independence in stroke survivors with neglect.

## POSTER ABSTRACTS

All Posters Will Be Presented From 12:30 pm to 2 pm in the Lilyan E. White Study Centre on the 4<sup>th</sup> floor.

### Ahmed, Mohammed Ali

**POSTER TITLE:** Age-Related Changes in Olfactory Discrimination and Reversal Learning Performance in Neurexin1 +/- Transgenic Mice

**AUTHORS:** Mohammed Ali Ahmed, Simone Shaw-Litzgus, Wyatt Ortibus, and Richard E. Brown

**ABSTRACT:** Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder marked by challenges in social communication, restricted interests, and repetitive behaviours. Nrnx1 is a presynaptic cell adhesion protein that interacts with postsynaptic neuroligins to regulate synaptic signalling. Nrnx1 is vital for learning and memory and has been associated with ASD. Nrnx1+/- transgenic mice have a knockdown of the  $\alpha$ ,  $\beta$ , and  $\gamma$  isoforms, leading to impaired synaptic transmission. Our study examined whether age (young vs. old) and sex (male vs. female) affected performance of Nrnx1+/- mice in olfactory discrimination and reversal learning tasks compared to C57BL/6J wildtype controls. Results showed that old mice had a significantly lower completion rate in the learning tasks, compared to young mice. Among the old mice, Nrnx1+/- mice had a lower completion rate than wildtype mice. The old mice had a significantly higher number of errors to the 85% criterion than the young mice in an initial discrimination task. These results suggest that as mice age, they face impairments in cognitive function and flexibility, indicating that this procedure may be used as a test for novel drugs which might increase cognitive function during aging.

### Atwell, Isabel

**POSTER TITLE:** Does Inhibiting Parvalbumin-Expressing Interneurons in Visual Cortex Alter Performance on a Perceptual Contrast Detection Task in Mice?

**AUTHORS:** Isabel F. Atwell, Nicole M. Michaud, Isabel F. Atwell, Daisy J. Gorski, Mark R. ten Haaf, Laura C. Rezende, Prem K. Singh, Lukas J. Zeisberger, Nathan A. Crowder

**ABSTRACT:** The application of optogenetic techniques in mouse primary visual cortex (V1) has allowed us to probe the roles for specific neuron subtypes within the neural circuits underlying sensory processing. Past research has identified inhibitory interneurons expressing parvalbumin (Pvalb+), somatostatin (SOM+) and vasoactive intestinal peptide (VIP+) as members of an inhibitory neural circuit that modulates excitatory pyramidal cell activity. We recently found that SOM+ inactivation alters perceptual contrast detection in a manner consistent with a visual phosphene effect, likely caused by release from inhibition of pyramidal cells. In this study we investigate whether Pvalb+ inactivation produced similar perceptual deficits using the same perceptual task. We show for the first time that Pvalb+ inactivation impairs performance on a contrast detection task.

### Cairae, Varun

**POSTER TITLE:** Characterizing Problematic Internet and Social Media Use Among Young Adults

**AUTHORS:** Simon B. Sherry; Aislin R. Mushquash; Christopher J. Mushquash; Daniel S. McGrath; Sherry H. Stewart

**ABSTRACT:** The Substance Use Risk Profile Scale (SURPS) is a measure that is used to select youth for personality-targeted



interventions (i.e., PreVenture) that have been found to prevent or reduce risky substance use in young people. We tested whether the SURPS impulsivity subscale would predict changes in binge drinking and binge eating over time. Undergraduates (N = 410, M age = 20.7 years, SD = 5.5, women n = 307, men, n = 103) from two universities completed a longitudinal study with two weeks separating baseline and follow-up. At baseline, participants completed the SURPS impulsivity subscale and binge drinking and binge eating measures. At follow up, participants completed the measures again. Linear regression analyses showed that SURPS impulsivity ( $B = .12$ ,  $p = .02$ ) significantly predicted binge drinking and binge eating at follow up. Findings reveal that the SURPS impulsivity subscale is a useful tool in predicting increases in binge behaviour.

### Dockrill, Mya

**POSTER TITLE:** Sleep Disparities in Canadian Children under 18 Years: A Scoping Review

**AUTHORS:** Mya Dockrill (Dalhousie University), Brendan Flores (Universite of Montreal), Youssouf Aidara (Universite of Montreal), Dr. Penny Corkum (Dalhousie University), Dr. Guido Simonelli (Universite of Montreal)

**ABSTRACT:** Insufficient sleep is prevalent in children and has many adverse health outcomes. Children facing health inequities (e.g., from a low socioeconomic background, or a racial/ethnic minority) have more sleep difficulties and are less likely to receive treatment. There is a lack of research investigating pediatric sleep and health inequities, particularly in Canada. Thus, the objective was to summarize the extant literature in this area. A list of health inequities was developed following the Canadian Institute of Health Research EDI dimensions and in consultation with the Canadian Sleep Research Consortium. Four key electronic databases (PubMed, EMBASE, PsycInfo, CINAHL) were searched. Studies were included if they were 1) in English or French, 2) original studies (e.g., empirical data collected), 3) conducted in Canada, 4) with children under 18 years, 5) population-based studies, and 6) reported on sleep outcomes and health inequities. Three reviewers screened 15,448 abstracts and titles resulting in 357 articles full-text articles. After screening, 25 studies were included in this review, focusing on sleep in the context of low socioeconomic status (18 studies), gender/sex (17 studies), family composition/caretaking (9 studies), housing/neighbourhood characteristics (8 studies), race/ethnicity (8 studies), immigrants/refugees (3 studies), and language (1 study). Of note, no distinction was made between sex and gender in the literature. Much of the research has been done with Caucasian, adolescents and school-aged children, in Quebec or Ontario. Generally, it appears that sleep health and disordered sleep are negatively affected by low socioeconomic status, poor housing/neighbourhood conditions, and in non-Caucasian children. The findings are varied in terms of sex differences with most studies suggesting no differences (76%) and some suggesting differences between boys and girls with 67% of the literature suggesting that boys sleep better. There is no clear link between sleep disturbances and family composition and caretaking. This is the first review to investigate the impact of health inequities on sleep in Canadian children. Identifying and understanding this link will inform the modification and development of sleep interventions to better meet the needs of these diverse individuals.

### Drew, Ben

**POSTER TITLE:** The Canadian Settler Guilt Scale: Measuring Settler Guilt in a Canadian Population

**AUTHORS:** Ben Drew & Chris Moore

**ABSTRACT:** Indigenous peoples in Canada have endured significant negative impacts stemming from the colonization of North America in the 1700s and cultural assimilation efforts by European colonizers. Research in Australia has demonstrated that some individuals of European ancestry feel guilt about the injustices and maltreatment that Indigenous populations have endured. This type of guilt, referred to as settler guilt, may be an important avenue towards reconciliation efforts between Indigenous and non-Indigenous populations. However, no research to our knowledge exists on the topic of settler guilt in Canada. To address this issue, the newly designed Canadian Settler Guilt Scale was administered to a sample of undergraduate university students at Dalhousie University to measure settler guilt in a Canadian population. The scale demonstrated strong construct validity and internal consistency through exploratory factor analysis and Cronbach's alpha calculations, and differences in settler guilt scores were observed across various demographic variables providing a strong foundation for future research on the topic.



## Faustova, Ana

**POSTER TITLE:** Seizure Symptoms in Neurexin-1 Knockdown and “Rescue” Mice with Slice Site 5 Modification: Determining the Relationship of Age, Genotype and Sex Using the Brown Lab Scale and KCNQ-1 Expression

**AUTHORS:** Ana Faustova, Wyatt Ortibus, Mohammed Ali Ahmed, Salma Ismail, Ian Weaver, and Richard E. Brown

**ABSTRACT:** The Neurexins (NRXNs) are a family of membrane protein synaptic organizers encoded by the *Nrxn* genes. *Nrxn*-1 mutations are associated with Autism Spectrum Disorder and Schizophrenia. We recorded seizures in male and female mouse models with a knockout of one *Nrxn*1 gene (*Nrxn*1<sup>+/-</sup> mice), wildtype C57Bl/6J controls (*Nrxn*1<sup>+/+</sup>), a\_n\_d\_  
\_N\_r\_x\_n\_1\_“r\_e\_s\_c\_u\_e”\_m\_o\_d\_e\_l\_w\_i\_t\_h\_a\_s\_p\_l\_i\_c\_e\_s\_i\_t\_e\_5\_a\_l\_t\_e\_r\_a\_t\_i\_o\_n\_ (DS5<sup>-/-</sup> and DS5/DS5 mice) (Lu, H., et al., 2023. **Cell Reports**, 42: 112714; Lu, H., et al. **Molecular Psychiatry**, 15 April 2025). The DS5 alteration increases NRXN-1 interaction with presynaptic ligands, counteracting the knockdown behavioral phenotype. Mice carrying the DS5 recovery were more likely to exhibit age-related seizures compared to *Nrxn* <sup>+/+</sup> and *Nrxn* <sup>+/-</sup> mice (48% DS5, 31% *Nrxn*1<sup>+/-</sup>, 25% *Nrxn*1<sup>+/+</sup>). There were no sex differences, but seizure risk increased in mice with age (17.5 months). No Mendelian heritability pattern was uncovered. The epigenetic analysis showed a complex interaction between seizures, KCNQ-1, and genotype. This mouse model therefore may be important for the study of age-related spontaneous seizures.

## Fernandes, Oswin

**POSTER TITLE:** Personality and Emotional Expression Across Languages: Comparing French and English Emerging Adults During COVID-19

**AUTHORS:** Fernandes, O., Snooks, T., Stewart, S. H., Yunus, F., Moore, M., Keough, M., Krank, M., Thompson, K., & Conrod, P.

**ABSTRACT:** This study examined how personality traits influence emotional expression in written narratives about COVID-19 pandemic impacts among French-Canadian undergraduates and whether these associations differ from those previously documented in English Canadian undergraduates. Drawing on Ekman’s Basic Emotions Theory and Plutchik’s Emotion Wheel, which posit universal emotional responses, we hypothesized that associations between personality traits and emotional word use would replicate cross-linguistically. Participants (N=1185; ages 18-25) from 5 Canadian universities (1 French-speaking; N=122) completed a free-response prompt describing the impact of the COVID-19 pandemic. Using the Substance Use Risk Profile Scale, we explored relationships between four traits, Anxiety Sensitivity, Hopelessness, Impulsivity, and Sensation Seeking—and emotional word use in French narratives, examined using the Linguistic Inquiry and Word Count tool. Data were analyzed using negative binomial regression and compared to Moore et al. (2025) findings from English-speaking students. Anxiety Sensitivity was associated with increased anxiety word use, and Sensation Seeking with greater anger word use, across both language groups. However, cross-linguistic differences emerged: French participants with high hopelessness expressed more anger, whereas their English counterparts expressed more sadness and fewer positive words. French Impulsivity individuals used fewer anxiety words, while English impulsive participants expressed more anger and sadness. Additionally, French sensation seekers used more sadness and general negative emotion words, while their English counterparts expressed more positive emotions. Results highlight both universal and language-specific patterns in personality-linked emotional expression and identify the types of emotional experiences that should be targeted in personality-matched mental health interventions for French undergraduates.

## Feltmate, Brett

**POSTER TITLE:** Is the Nature of the Vigilance Decrement Affected by Target Probability?

**AUTHORS:** Brett Feltmate, Paige Dobson, Mia Bartlett, Alyssa Dober, Elise Windybank, Ray Klein

**ABSTRACT:** In tasks requiring vigilant detection of rare and noisy signals, humans ubiquitously commit more errors as time progresses. A consistent theme throughout the literature is how best to characterize this "vigilance" decrement: is it a

decline in one's ability to discriminate signals, or a conservative shift in their criterion to respond? Given that both accounts have empirical support, it is more likely that task factors moderate how the vigilance decrement is felt. We explored if/how people's ability to discriminate signals, and their bias in categorizing them as such, are differentially sensitive to target likelihood. Participants signalled whether a central line was sufficiently offset from those flanking it, which occurred on (between-Ss) 25/50/75% of trials (900/ participant). Results hint toward discriminability and response biases dissociating over time, as target likelihood approaches either limit.

### Haggett, Sarah

**POSTER TITLE:** Heightened Error-Monitoring Following Childhood Adversity: An ERP Investigation of the ERN

**AUTHORS:** Sarah M. Haggett, Christine L. Lackner

**ABSTRACT:** Adverse childhood experiences (ACEs) are linked to disruptions in brain regions involved in self-regulation and performance monitoring. The error-related negativity (ERN) is a neural marker of error awareness. Although self-regulation is associated with ERN amplitude, findings on the ACE-ERN relationship are inconsistent, possibly due to simplified ACE measures. This study is the first to examine four ACE dimensions – type, timing, extent, and subjective evaluation – in relation to ERN. One hundred young adults will complete ACE questionnaires and perform four Go/NoGo tasks during EEG recording. Tasks will vary in emotional salience (neutral, emotional) and sensory modality (auditory, visual). We hypothesize that ACE extent and subjective severity will negatively correlate with ERN across tasks. The timing–ERN relationship will depend on age at exposure. ACE type effects will depend on task salience, with greater effects during emotional tasks, and on sensory modality, with more negative ERNs when modality aligns with experienced ACEs.

### Imam, Aurthy

**POSTER TITLE:** Leveraging Speech Analysis for Cognitive Load Measurement in Young Adults: A Sex-Based Comparative Study

**AUTHORS:** Aurthy Imam

**ABSTRACT:** This study explored how cognitive load impacts speech production and whether sex or baseline fatigue moderates this relationship. Thirty young adults completed low and high cognitive load speech tasks, which involved spontaneous speech and dual-task conditions with arithmetic and verbal fluency challenges. Acoustic (e.g., speech rate, pitch) and disfluency features (e.g., filled/silent pauses) were analyzed using PRAAT and LIWC, while perceived load was assessed with the NASA-TLX. Results showed significant decreases in speech rate and increases in filled pauses under higher cognitive load. Participants with higher fatigue scores exhibited greater changes in speech rate and pitch modulation. These findings support the use of speech as a non-invasive marker of cognitive load and point toward potential applications in workload monitoring across clinical, educational, and occupational settings.

### Inness, Bronwyn

**POSTER TITLE:** Longitudinal Effects of Parental Smartphone Usage on Child Socioemotional Wellbeing Proposal

**AUTHORS:** Inness, B., Montalvo, A., Mason, L., Khoury, J., Stewart, S. H., & Meier, S.

**ABSTRACT:** Parental connection and responsiveness have been well-established to play a central role in children's social wellbeing and emotional health. However, integrating smartphone technology into daily life has caused concern about how these devices might significantly interrupt parental responsiveness. Research has found differences in parenting practices due to smartphone use, consequently impacting children's socioemotional health. There are a variety of mechanisms that might explain these impacts, however, the details have yet to be identified. This study will investigate the two primary postulated mechanisms: "technoference", the interruption of interactions with children by a smartphone, and "immersion", the absorption of parents' attention by the smartphone. Using a multimodal longitudinal design, 150 parent dyads with 12 to 18-month-olds will complete online questionnaires, download a mobile sensing app, and be observed interacting with their child. It is hypothesized that heavy parental smartphone use, objectively recorded via

mobile sensing, will predict poor parental responsiveness and child socioemotional development over time. Technoference and immersion are expected to mediate the association between smartphone use, parental responsiveness, and child socioemotional wellbeing. This study aims to inform parents, policymakers, clinicians, and researchers on how and when parental smartphone use impacts child socioemotional development.

### Ismail, Salma

**POSTER TITLE:** Effects of Age on Motor Coordination and Motor Learning in Mouse Models of Autism

**AUTHORS:** Salma Ismail, William Grendon and Richard E. Brown

**ABSTRACT:** Autism spectrum disorder (ASD) is a neurodevelopmental disorder that impairs social-emotional behaviour, motor coordination, visual-spatial learning and memory. Approximately 2.5% of the human adult population has ASD and adults with ASD have a reduced lifespan, increased frailty and motor gait impairments. The present study assessed motor coordination and motor learning using the Rotarod in 21-month-old male and female *Nrxn1*<sup>+/-</sup> transgenic mice; wildtype (C57BL/6J) controls; knockdown rescue  $\Delta S5$ - ; and double rescue  $\Delta S5/\Delta S5$  mice. The mice were placed in one of the 4 compartments of the Rotarod, which accelerated to a maximum of 36 rpm and the latency to fall was recorded. There were 6 trials per day over 7 days and body weight was taken prior to testing each day. Performance improved for all mice over days, however, there was a significant negative correlation between motor scores and body weight ( $r = -0.43$ ,  $df = 97$ ,  $p < 0.001$ ) with lighter mice performing better than heavier mice. Females performed better than males, but improvement in performance over days was greater in males. Female *Ds5*<sup>-/-</sup> mice performed better than female WT, *Nrxn1*<sup>+/-</sup> and *DS5/DS5* mice, but there were no genotype differences in male mice. At 2 and 9 month of age, female mice outperformed males but there were no differences between *Nrxn1*<sup>+/-</sup> and WT mice. These results indicate that there are no age-related deficits in motor learning and coordination in these autistic mouse models.

### Jones, Wes

**POSTER TITLE:** Investigation of Astrocyte Activation in the Visual Cortex Following Monocular Retinal Inactivation

**AUTHORS:** Wes Jones & Kevin Duffy

**ABSTRACT:** Early discordant vision can cause amblyopia, a lasting impairment usually affecting one eye. In animal models of amblyopia, inactivating the dominant retina with tetrodotoxin elicits better recovery than conventional patching treatment. Despite inactivating the retina, connectivity within the visual system is preserved. Astrocyte activation may support this preservation through homeostatic mechanisms. Our lab previously found elevated glial fibrillary acidic protein (GFAP), a marker of astrocyte activation, in layers of the lateral geniculate nucleus serving the inactivated eye. The current study investigated whether astrocyte activation also occurs in the visual cortex (V1). We compared GFAP labeling between control animals and those subjected to 10 days of MI. While young animals showed no difference, older animals exhibited significantly reduced GFAP after MI. We propose that this decrease in GFAP reflects an upstream synaptic compensation in response to inactivation-induced changes elicited within the lateral geniculate nucleus.

### Longino, Kirsty

**POSTER TITLE:** Exploring How to Assess Orthographic Learning in Grade 1

**AUTHORS:** Longino, Kirsty (presenter); Dr. Giazitzidou, Sophia; Hoferek, Katie; Dr. Deacon, Hélène.

**ABSTRACT:** Orthographic learning refers to children's skill in learning the spelling of novel words through their independent reading, and it has been described as central to reading development. Measuring this skill in young readers is challenging, as it requires—by definition—independent reading. The aim of this pilot study was to develop an appropriate task to assess orthographic learning in English-speaking children in Grade 1. As in prior studies, we asked students to read 12 stories that contained made-up inventions. The first method required students to read independently. 25% of students could not accomplish this due to limited reading ability and loss of assent associated with reading difficulty. The second method shifted to shared reading for those who did not have the necessary reading skills, but had them decode

the target words independently. This enabled all students to participate, with only one case of lost assent. In both methods, students were assessed immediately following the stories, and the following day, on the spelling and meaning of the inventions. Results showed that only independent reading demonstrated reliability (Cronbach's  $\alpha > .7$ ) while highlighting the importance of multiple methods to ensure full participation.

**MacDougall, Erin**

**POSTER TITLE:** Does Hazardous Drinking in First Year University Students Predict Academic Performance into Their Fourth Year? A Study of Students with and Without a History of Reading Difficulties

**AUTHORS:** MacDougall, Erin (presenter); Elgendi, Mariam; Dr. Stewart, Sherry; Dr. Deacon, Hélène

**ABSTRACT: Background:** Both problematic drinking and a history of reading difficulties have been associated with academic challenges in university students. Here we build on these two bodies of work to examine the longitudinal effects of alcohol use on academic performance in a study comparing students with and without a self-reported history of reading difficulties (HRD and NRD, respectively). **Methods:** We report on a four-year longitudinal study following 93 university students' (46 HRD, 47 NRD) GPA in Year 1 and Year 4. In Year 1, students completed online questionnaires assessing both hazardous drinking and a self-reported history of reading difficulties. **Results:** Linear regression analyses showed that hazardous drinking was negatively associated with changes in GPA from Year 1 to Year 4, for both students with a without a history of reading difficulties. Follow-up analyses identified drinking quantity as the primary contributor. **Discussion:** Overall, findings emphasize the need for university-based interventions targeting drinking behaviour to support academic outcomes, regardless of reading history.

## Savoie, Samantha

**POSTER TITLE:** Decision Making in Mice: A Detailed Analysis of Behaviour in Mice Navigating a Visual Discrimination Task

**AUTHORS:** Samantha Savoie, Richard Brown

**ABSTRACT:** Behavioural studies using mouse models commonly involve visual stimuli, even though mice are not known to be visually guided animals. Wong & Brown 2006, *Genes Brain and Behaviour* 5, 389-403 examined the ability of different strains of mice to discriminate between visual stimuli in a forced-choice swim task. How mice make decisions while learning this type of task remains under-researched. We analyzed swim paths to determine decision-making tactics in the visual water box. We recorded 1) correct decision-making, 2) errors, 3) early error correction, 4) late error correction, 5) vicarious trial and error (VTE), and 6) thigmotaxis in 12 strains of mice with varying visual abilities. Using digitized footage, we coded the swim paths and compared the time taken to complete correct paths between visual ability groups. Mice were scored based on their search strategies and the time taken to complete the correct path. We found that visually impaired mice relied more on non-visual search strategies and made more error paths than those with good vision, who were likelier to use VTE and early error corrections. These findings show that mice with differing visual abilities use different search strategies. Mice with good vision showed longer times for VTE and error paths, which suggests more deliberative strategies. Learning trends varied by visual ability across trials. Despite limitations due to video quality and sample size, these results provide more information on how search strategies of mice differ based on visual acuity.

## Sozuer, Can

**POSTER TITLE:** Compartmental Modeling of the Whole-Cell Activity of Wake-Promoting Orexin Neurons in Response to the Sleep Deprivation–Induced Astrocytic Inhibition of Their Excitatory Inputs

**AUTHORS:** Can Sozuer, Vikas Pandey, Yoshiyuki Kubota, Kazue Semba

**ABSTRACT:** Electron microscopy and electrophysiology revealed that six-hour sleep deprivation induces astrocyte withdrawal from perisynaptic regions and presynaptic inhibition at excitatory synapses onto wake-promoting orexin neurons (Briggs et al., 2018, JNeurosci; Semba et al., 2023, SfN). Computational synaptic modeling demonstrated that astrocyte withdrawal drives presynaptic inhibition by modulating glutamate uptake. Although these alterations may

homeostatically and substantially attenuate orexin neuron firing, their functional consequences remain unclear. To bridge synaptic changes and neuronal output, we developed the first detailed conductance-based compartmental model of an orexin neuron in our NEURON simulation environment, replicating D- and H-type firing from intrinsic membrane channel and receptor mechanisms. We stimulated this model with excitatory post-synaptic currents reflecting varying degrees of astrocyte withdrawal—holding other parameters constant and incorporating additional sleep-deprivation-induced physiological changes. By linking astrocyte withdrawal to orexin neuron firing, we extend synaptic-level observations into a detailed multiscale integrative framework, illuminating how nanoscale synaptic alterations may potentially impact wakefulness.

### ten Haaf, Mark

**POSTER TITLE:** Optogenetic Activation of Somatostatin Expressing Interneurons in V1 Impairs Size Perception in Transgenic Mice

**AUTHORS:** ten Haaf, M.R., Michaud, N.M., Atwell, I.F., Dang, L.T.J., Gorski, D.J., Rezende, L.C., Singh, P.K., Zeisberger, L.J., Crowder, N.A.

**ABSTRACT:** Mice are a valuable model in neuroscience research as they can be used with a variety of genetic tools to link neural circuits and behavior. In this study, we use optogenetics, where specific neural subtypes can be made to express light sensitive ion channels that can modulate neural activity, to investigate the role of Somatostatin expressing interneurons in visual perception. Specifically, we used transgenic mice expressing Channelrhodopsin-2 channels to depolarize somatostatin expressing interneurons in the primary visual cortex while mice completed a two-alternative-forced-choice size detection task. Prior findings from our lab demonstrated that inhibiting somatostatin expressing interneurons impaired size perception, so we hypothesized that activating these neurons might enhance size perception. Contrary to our hypothesis the activation did not enhance performance so here we explore this effect and its implications on the role of cortical inhibition in sensory processing.

### Whitfield, Caden

**POSTER TITLE:** The Impact of Communal Nesting on Adolescent Stress-Induced Intergenerational Effects in Male and Female Mice

**AUTHORS:** Caden Whitfield, Tasia MacLean, Ewa Adedeeji, Mackenzie R. Hartley, Tamara B. Franklin

**ABSTRACT:** Chronic stress during the adolescent period has proven to be impactful to mental health and behaviour throughout adulthood. Previous research in our lab has demonstrated that chronic adolescent stress in mice induced behaviour similar to postpartum depression, with the dams showing reduced motivation and providing less maternal care to their pups. Altered behaviour states during the postpartum period impact the pups' behaviour in adulthood, with pups demonstrating increased anxiety-like behaviour. The current research focuses on assessing interventions to help counter these intergenerational effects by investigating the potential of communal nesting as a model for social support in the postpartum period. F0 generation mice undergo a predator stress protocol daily over the course of 12 days during the early adolescent period to mimic the experiences of chronic adolescent psychological stress. These mice are bred in same-treatment pairings and once pups are born, are either placed alone in standard housing or housing with another dam and their pups. As adults, the offspring were tested in a battery of tests aimed at assessing exploration, anxiety-like behaviours and depressive-like behaviours: olfactory habituation/dishabituation, open field, sucrose splash test and the elevated plus maze. This study is ongoing, however preliminary results from this behavioural testing suggest that communal nesting can modulate some, but not all of the effects of ancestral stress.

## Whitman, Maura

**POSTER TITLE:** Comparing Attitudes Toward Diverse and Homogeneous Groups Varying in Race, Accent, and Sex/Gender Among Canadian Primary-School-Aged Children

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**ABSTRACT:** Children in Canada are learning language and interacting with others in far more socially (e.g., race and ethnicity) and linguistically (e.g., languages and accents) diverse environments than ever before. We explored children's attitudes toward diverse and homogeneous social groups using ecologically valid methods that reflect the real-world contexts in which diversity is experienced. Data collection is ongoing. Fifty-five Canadian children, aged 5 to 8 years (males  $n = 30$ , females  $n = 25$ ,  $M$  age = 7.8 years), completed three blocks of trials varying by social dimension: sex/gender, race, or accent. On each trial, participants were presented with cartoon images of groups that differed according to the social dimension being evaluated within each block: one group was homogeneous, and the other was diverse. Participants were then asked to rate how much they liked each group using a Likert-response scale consisting of emoticons. Wilcoxon Signed-Rank tests will be used to determine whether responses differ significantly between the two types of stimuli (e.g., diverse vs. homogenous groups) and across age groups. Findings can have implications for educational practice, policy-making, and early interventions that promote acceptance and support for diversity.

## Yildirim, Deniz

**POSTER TITLE:** Psychopathic Traits and Cooperation: The Role of Punishment Risk in Social Decision-Making

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**ABSTRACT:** Cooperation is integral to a functioning society, but people higher in psychopathic traits (e.g., showing callous affect, manipulateness, impulsivity, antisocial traits) may be unwilling to cooperate and consistently violate social norms. The current study investigated the extent to which cooperative decision-making is influenced by psychopathy scores and punishment risk, as measured through three hypothetical Public Goods Game (PGG) scenarios with varying certainties of (monetary) punishment risk (i.e., low-, moderate-, and high-risk). The sample consisted of 455 undergraduate students, recruited through SONA. Participants completed the three hypothetical PGG scenarios and a multifaceted psychopathy measure: the Self-Report Psychopathy Scale – Short Form (SRP-SF). As hypothesized, participants contributed more points in conditions that offered increased risk of punishment. However, total psychopathy scores did not significantly predict lower contributions averaged across the three PGG conditions. Higher psychopathy scores were also not associated with the difference in contributions between low- and high-risk conditions, nor between the low- and moderate-risk conditions. Nonetheless, higher psychopathy scores, specifically on the Erratic Lifestyle facet, were correlated with lower contributions in the moderate-risk condition. Our findings highlight the complex nature of psychopathy and suggest that psychopathy does not uniformly predict cooperative behaviour but may be influenced by the likelihood of punishment as well as certain psychopathy facets.